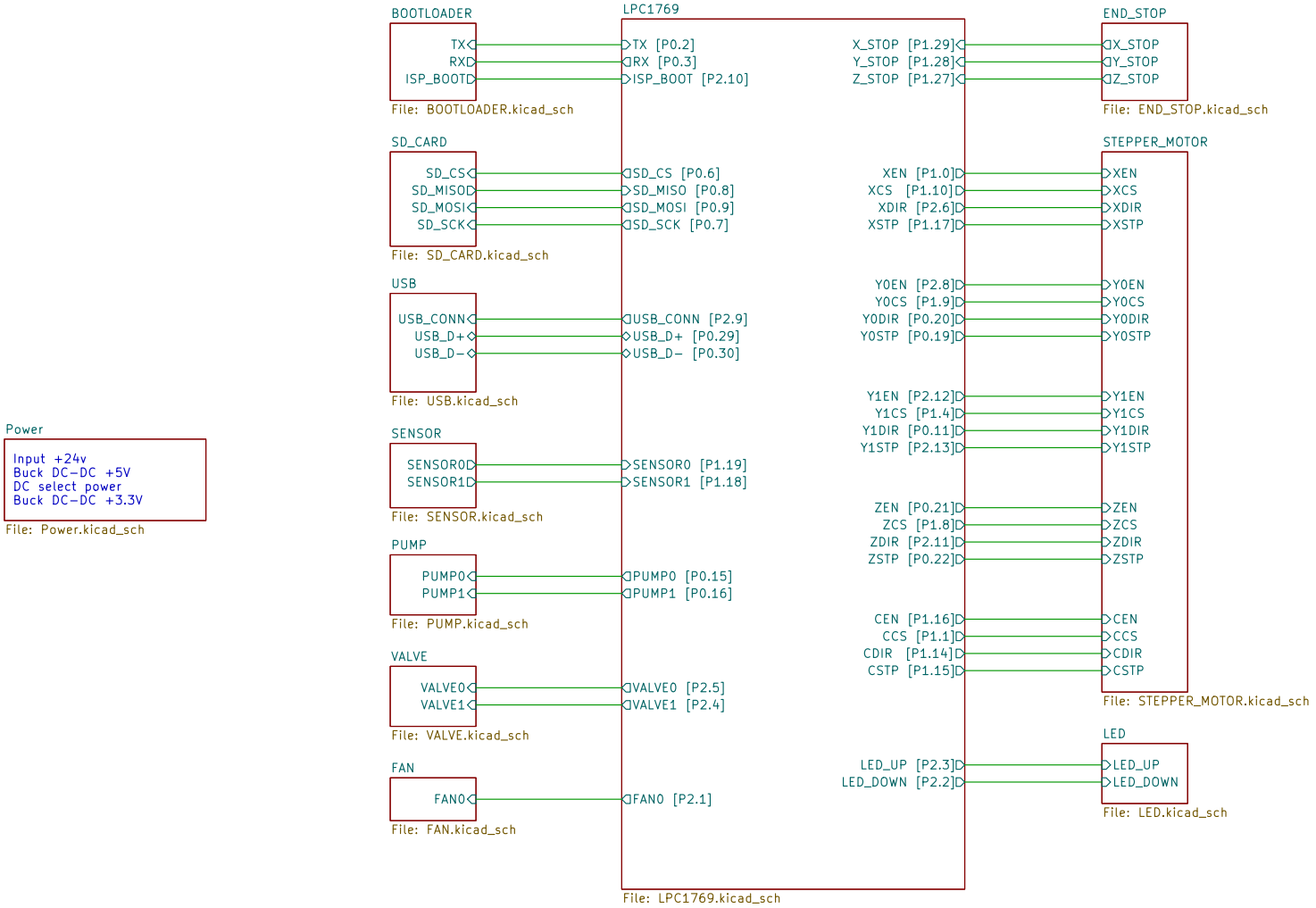
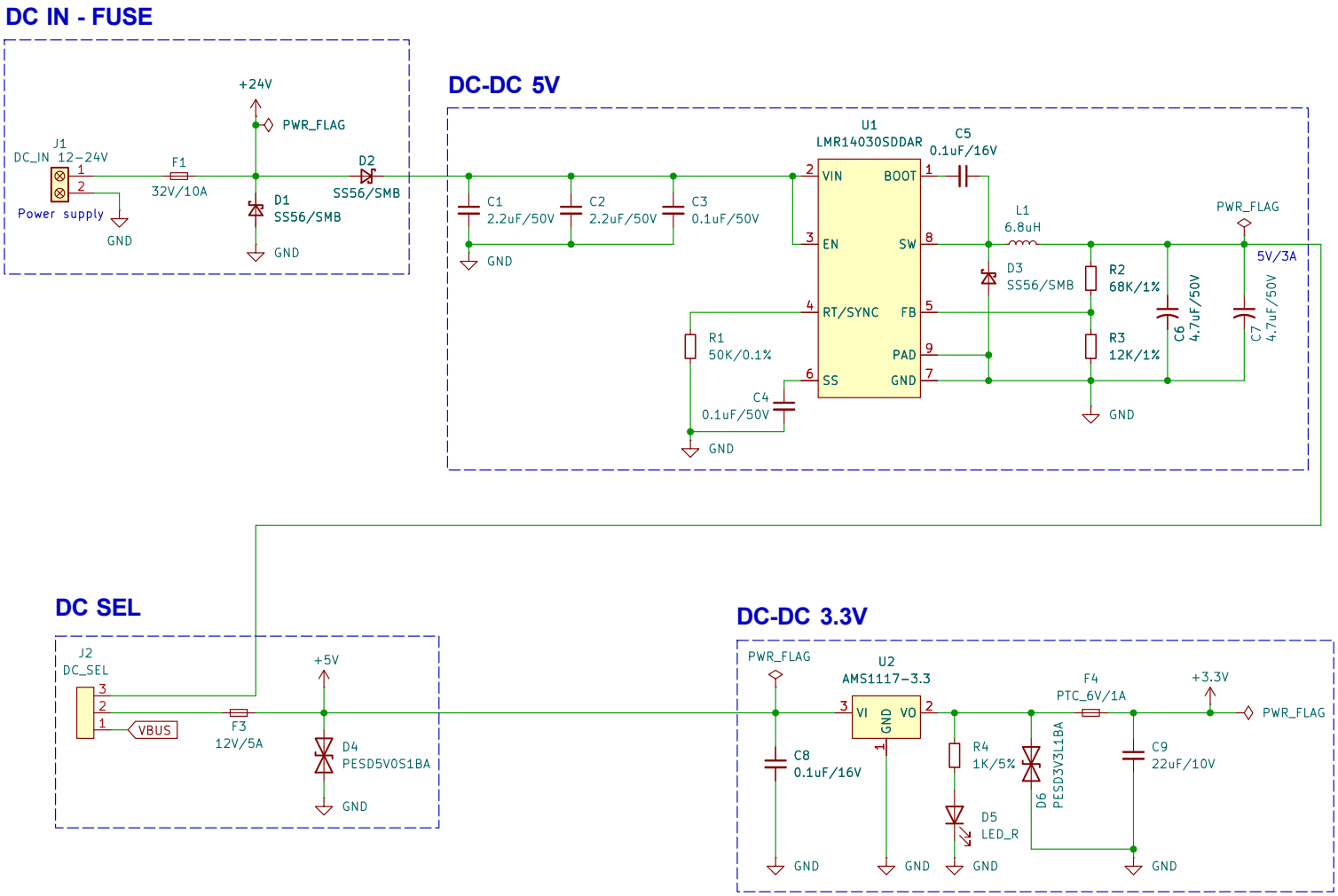
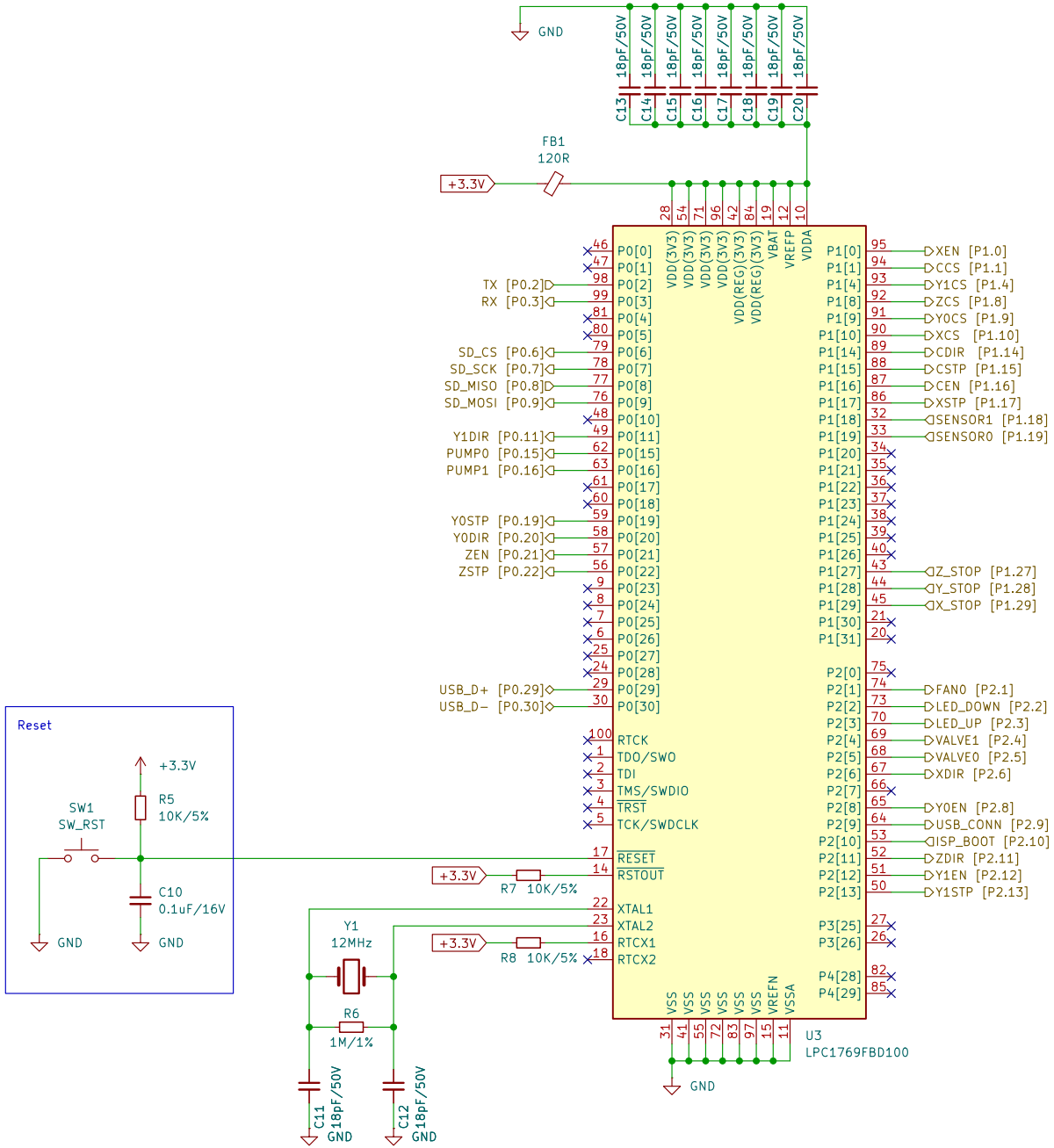
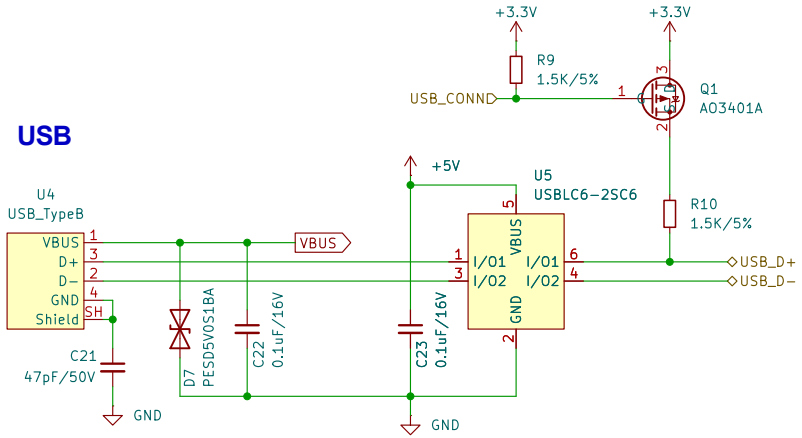
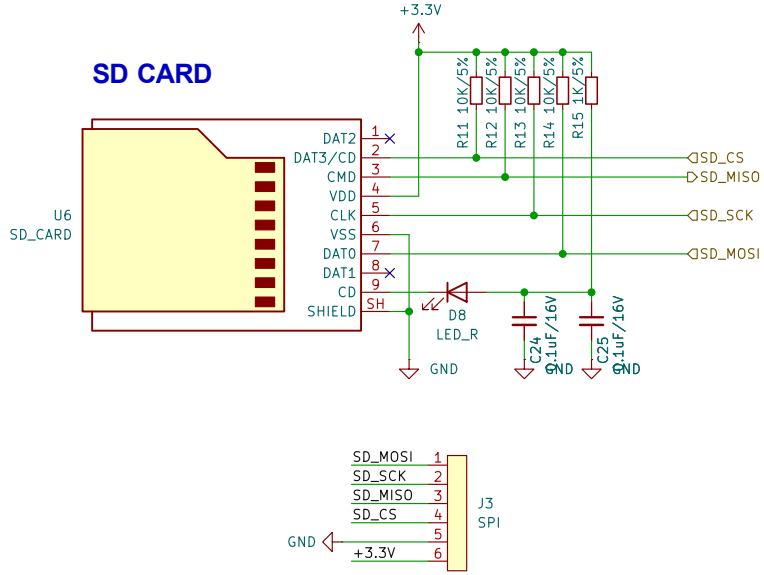

Power
Input +24v
Buck DC-DC +5V
DC select power
Buck DC-DC +3.3V
File: Power.kicad_sch
BOOTLOADER
TX
RX
ISP_BOOT
File: BOOTLOADER.kicad_sch
SD_CARD
SD_CS
SD_MISO
SD_MOSI
SD_SCK
File: SD_CARD.kicad_sch
USB
USB_CONN
USB_D+
USB_D-
File: USB.kicad_sch
SENSOR
SENSOR0
SENSOR1
File: SENSOR.kicad_sch
PUMP
PUMP0
PUMP1
File: PUMP.kicad_sch
VALVE
VALVE0
VALVE1
File: VALVE.kicad_sch
FAN
FAN0
File: FAN.kicad_sch
LPC1769
TX [P0.2]
RX [P0.3]
ISP_BOOT [P2.10]
SD_CS [P0.6]
SD_MISO [P0.8]
SD_MOSI [P0.9]
SD_SCK [P0.7]
USB_CONN [P2.9]
USB_D+ [P0.29]
USB_D- [P0.30]
SENSOR0 [P1.19]
SENSOR1 [P1.18]
PUMP0 [P0.15]
PUMP1 [P0.16]
VALVE0 [P2.5]
VALVE1 [P2.4]
FAN0 [P2.1]
X_STOP [P1.29]
Y_STOP [P1.28]
Z_STOP [P1.27]
XEN [P1.0]
XCS [P1.10]
XDIR [P2.6]
XSTP [P1.17]
Y0EN [P2.8]
Y0CS [P1.9]
Y0DIR [P0.20]
Y0STP [P0.19]
Y1EN [P2.12]
Y1CS [P1.4]
Y1DIR [P0.11]
Y1STP [P2.13]
ZEN [P0.21]
ZCS [P1.8]
ZDIR [P2.11]
ZSTP [P0.22]
CEN [P1.16]
CCS [P1.1]
CDIR [P1.14]
CSTP [P1.15]
LED_UP [P2.3]
LED_DOWN [P2.2]
File: LPC1769.kicad_sch
END_STOP
X_STOP
Y_STOP
Z_STOP
File: END_STOP.kicad_sch
STEPPER_MOTOR
XEN
XCS
XDIR
XSTP
Y0EN
Y0CS
Y0DIR
Y0STP
Y1EN
Y1CS
Y1DIR
Y1STP
ZEN
ZCS
ZDIR
ZSTP
CEN
CCS
CDIR
CSTP
File: STEPPER_MOTOR.kicad_sch
LED
LED_UP
LED_DOWN
File: LED.kicad_sch

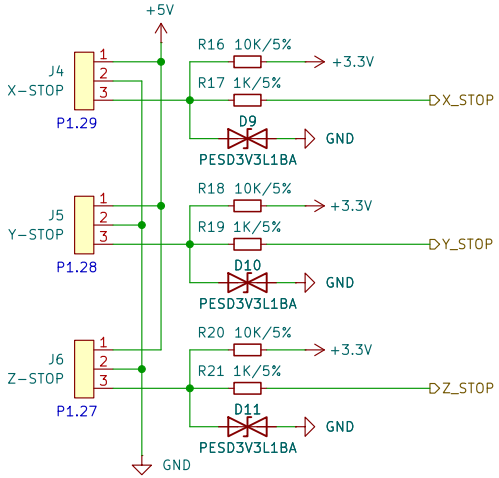

DC IN - FUSE
+24V
PWR_FLAG
J1
DC_IN 12-24V
1
2
Power supply
GND
F1
32V/10A
D1
SS56/SMB
D2
SS56/SMB
GND
DC-DC 5V
U1
LMR14030SDDAR
VIN
BOOT
EN
SW
RT/SYNC
FB
PAD
SS
GND
C1
2.2uF/50V
C2
2.2uF/50V
C3
0.1uF/50V
C4
0.1uF/50V
C5
0.1uF/16V
L1
6.8uH
D3
SS56/SMB
R1
50K/0.1%
R2
68K/1%
R3
12K/1%
C6
4.7uF/50V
C7
4.7uF/50V
PWR_FLAG
5V/3A
GND
DC SEL
J2
DC_SEL
3
2
1
VBUS
F3
12V/5A
+5V
D4
PESD5V0S1BA
GND
DC-DC 3.3V
PWR_FLAG
U2
AMS1117-3.3
VI
GND
VO
C8
0.1uF/16V
R4
1K/5%
D5
LED_R
D6
PESD3V3L1BA
F4
PTC_6V/1A
+3.3V
C9
22uF/10V
PWR_FLAG
GND

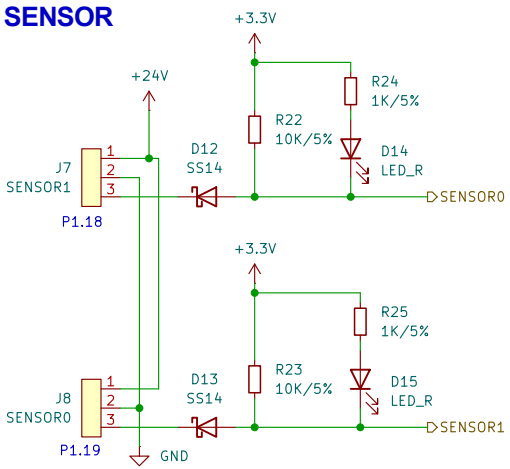

GND
18pF/50V
C13
C14
C15
C16
C17
C18
C19
C20
FB1
120R
+3.3V
U3
LPC1769FBD100
P0[0]
P0[1]
TX [P0.2]
RX [P0.3]
P0[4]
P0[5]
SD_CS [P0.6]
SD_SCK [P0.7]
SD_MISO [P0.8]
SD_MOSI [P0.9]
P0[10]
Y1DIR [P0.11]
PUMP0 [P0.15]
PUMP1 [P0.16]
P0[17]
P0[18]
Y0STP [P0.19]
Y0DIR [P0.20]
ZEN [P0.21]
ZSTP [P0.22]
P0[23]
P0[24]
P0[25]
P0[26]
P0[27]
P0[28]
USB_D+ [P0.29]
USB_D− [P0.30]
RTCK
TDO/SWO
TDI
TMS/SWDIO
TRST
TCK/SWDCLK
RESET
RSTOUT
XTAL1
XTAL2
RTCX1
RTCX2
VDD(3V3)
VDD(REG)(3V3)
VBAT
VREFP
VDDA
VSS
VREFN
VSSA
P1[0] XEN [P1.0]
P1[1] CCS [P1.1]
P1[4] Y1CS [P1.4]
P1[8] ZCS [P1.8]
P1[9] Y0CS [P1.9]
P1[10] XCS [P1.10]
P1[14] CDIR [P1.14]
P1[15] CSTP [P1.15]
P1[16] CEN [P1.16]
P1[17] XSTP [P1.17]
P1[18] SENSOR1 [P1.18]
P1[19] SENSOR0 [P1.19]
P1[20]
P1[21]
P1[22]
P1[23]
P1[24]
P1[25]
P1[26]
P1[27] Z_STOP [P1.27]
P1[28] Y_STOP [P1.28]
P1[29] X_STOP [P1.29]
P1[30]
P1[31]
P2[0]
P2[1] FAN0 [P2.1]
P2[2] LED_DOWN [P2.2]
P2[3] LED_UP [P2.3]
P2[4] VALVE1 [P2.4]
P2[5] VALVE0 [P2.5]
P2[6] XDIR [P2.6]
P2[7]
P2[8] Y0EN [P2.8]
P2[9] USB_CONN [P2.9]
P2[10] ISP_BOOT [P2.10]
P2[11] ZDIR [P2.11]
P2[12] Y1EN [P2.12]
P2[13] Y1STP [P2.13]
P3[25]
P3[26]
P4[28]
P4[29]
R7 10K/5%
R8 10K/5%
Y1
12MHz
R6
1M/1%
C11 18pF/50V
C12 18pF/50V
Reset
SW1
SW_RST
R5
10K/5%
C10
0.1uF/16V


USB
+3.3V
+3.3V
R9
1.5K/5%
USB_CONND
Q1
AO3401A
+5V
U5
USBLC6-2SC6
R10
1.5K/5%
U4
USB_TypeB
VBUS
D+
D-
GND
Shield
SH
VBUS
I/O1
I/O2
I/O1
I/O2
VBUS
GND
USB_D+
USB_D-
C21
47pF/50V
D7
PESD5V0S1BA
C22
0.1uF/16V
C23
0.1uF/16V
GND
GND

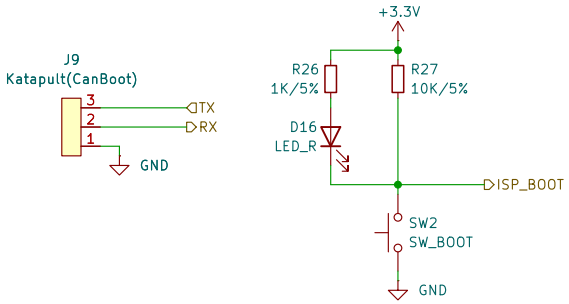

SD CARD
+3.3V
U6
SD_CARD
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
CD
SHIELD
1
2
3
4
5
6
7
8
9
SH
R11 10K/5%
R12 10K/5%
R13 10K/5%
R14 10K/5%
R15 1K/5%
SD_CS
SD_MISO
SD_SCK
SD_MOSI
D8
LED_R
C24 0.1uF/16V
C25 0.1uF/16V
GND
GND
GND
SD_MOSI
SD_SCK
SD_MISO
SD_CS
GND
+3.3V
1
2
3
4
5
6
J3
SPI

ENDSTOP

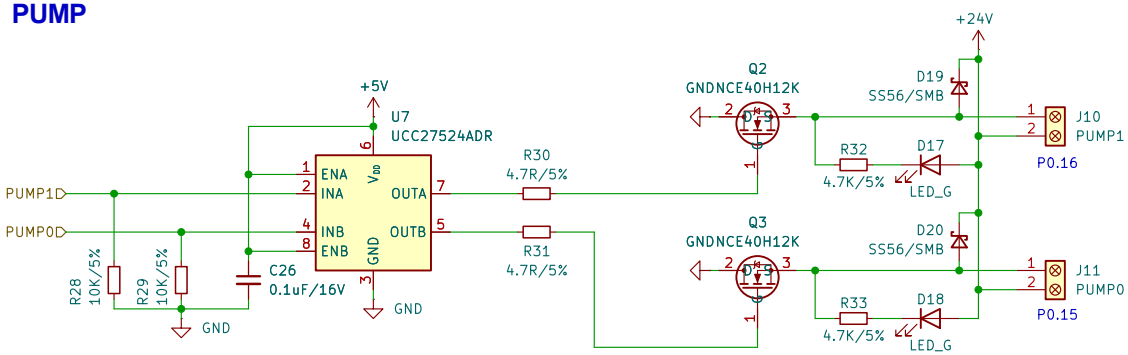

+5V
J4
X-STOP
P1.29
1
2
3
R16 10K/5%
+3.3V
R17 1K/5%
X_STOP
D9
PESD3V3L1BA
GND
J5
Y-STOP
P1.28
R18 10K/5%
+3.3V
R19 1K/5%
Y_STOP
D10
PESD3V3L1BA
GND
J6
Z-STOP
P1.27
R20 10K/5%
+3.3V
R21 1K/5%
Z_STOP
D11
PESD3V3L1BA
GND
GND

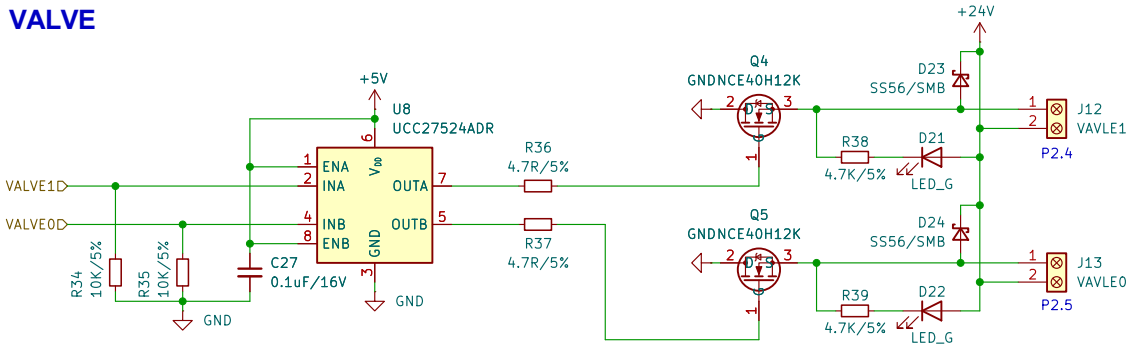

SENSOR
+3.3V
+24V
R24
1K/5%
R22
10K/5%
D12
SS14
D14
LED_R
J7
SENSOR1
1
2
3
P1.18
SENSOR0
+3.3V
R25
1K/5%
R23
10K/5%
D13
SS14
D15
LED_R
J8
SENSOR0
1
2
3
P1.19
GND
SENSOR1

BOOTLOADER

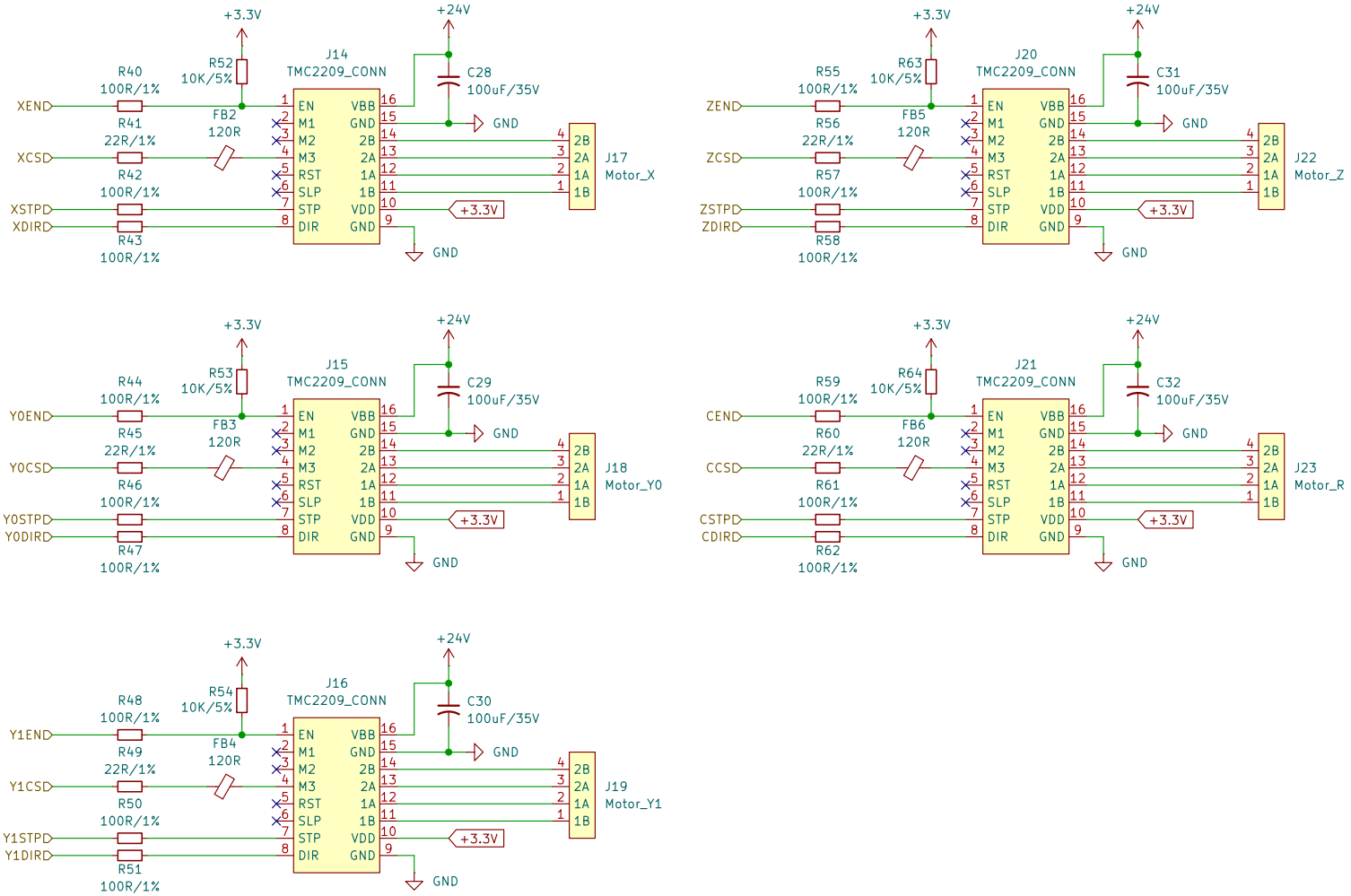

J9
Katapult(CanBoot)
3
2
1
TX
RX
GND
+3.3V
R26
1K/5%
R27
10K/5%
D16
LED_R
ISP_BOOT
SW2
SW_BOOT
GND

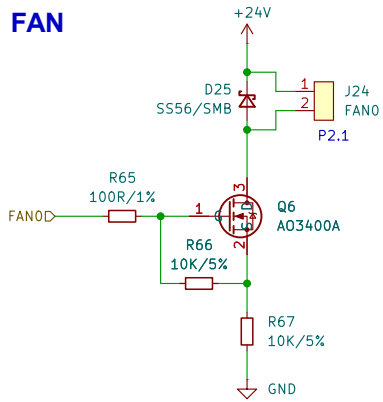
PUMP

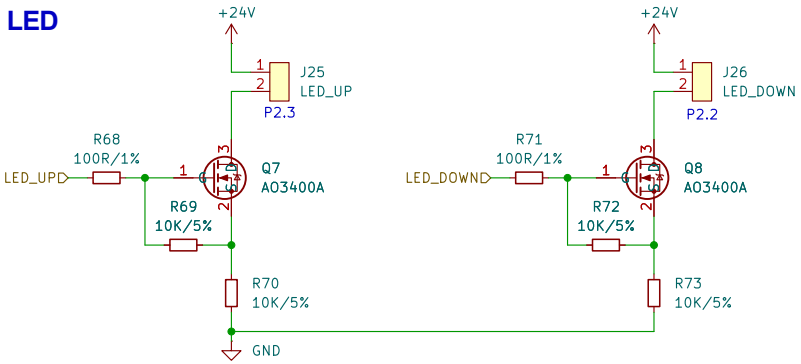

+5V
U7
UCC27524ADR
ENA
INA
INB
ENB
VDD
GND
OUTA
OUTB
PUMP1
PUMP0
R28
10K/5%
R29
10K/5%
C26
0.1uF/16V
GND
GND
R30
4.7R/5%
R31
4.7R/5%
+24V
Q2
GNDNCE40H12K
Q3
GNDNCE40H12K
D19
SS56/SMB
D20
SS56/SMB
R32
4.7K/5%
R33
4.7K/5%
D17
LED_G
D18
LED_G
J10
PUMP1
P0.16
J11
PUMP0
P0.15


VALVE
+5V
U8
UCC27524ADR
ENA
INA
INB
ENB
VDD
OUTA
OUTB
GND
VALVE1
VALVE0
R34
10K/5%
R35
10K/5%
C27
0.1uF/16V
R36
4.7R/5%
R37
4.7R/5%
+24V
Q4
GNDNCE40H12K
Q5
GNDNCE40H12K
D23
SS56/SMB
D24
SS56/SMB
R38
4.7K/5%
R39
4.7K/5%
D21
LED_G
D22
LED_G
J12
VAVLE1
P2.4
J13
VAVLE0
P2.5

STEPPER MOTOR




FAN
+24V
D25
SS56/SMB
1
2
J24
FAN0
P2.1
R65
100R/1%
FAN0
1
2
3
Q6
AO3400A
R66
10K/5%
R67
10K/5%
GND


LED
+24V
J25
LED_UP
P2.3
R68
100R/1%
LED_UP
R69
10K/5%
Q7
AO3400A
R70
10K/5%
GND
+24V
J26
LED_DOWN
P2.2
R71
100R/1%
LED_DOWN
R72
10K/5%
Q8
AO3400A
R73
10K/5%